MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, THYRISTORS (CONTROLLED RECTIFIERS), SILICON TYPES 2N1771A, 2N1772A, 2N1774A, 2N1776A, 2N1777A, TX2N1771A, TX2N1772A, TX2N1774A, TX2N1776A, TX2N1777A

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNPN silicon power, reverse-blocking triode, thyristors and is in accordance with MIL-S-19500 except as otherwise specified herein. The prefix "TX" is used on devices submitted to and passing the special process-conditioning, testing, and screening specified in 4.6 through 4.6.6.3.

1.2 Physical dimensions. See figure 1 (TO-64).

1.3 Characteristics common to all types.

	V_{FM}	I_{HOO}	V_{GT} 1/	I_{GT} 1/	t_{off} 1/	t_{on}	Critical dv/dt 1/
	(v)	(mAdc)	(Vdc)	(mAdc)	(μsec)	μsec	v/μsec
Minimum	---	---	0.2	---	---	---	5
Maximum	1.85	25.0	2.0	30.0	30	5	---

1/ T_C = -65 to +150° C.

1.4 Ratings. (See 6.2)

1.4.1 Ratings common to all types.

	I_o 1/ 3/ T_C = +105° C (180° cond. angle)	i_{FM} (surge) 2/ 1 cycle	V_{KGM}	T_{op}	T_{stg}
	(A)	(a)	v(pk)	(° C)	(° C)
Minimum	---	---	---	-65	-65
Maximum	4.7	60	10	+150	+150

1/ This average forward current is for a maximum case temperature of +105° C, and 180 electrical degrees of conduction. For other maximum operating conditions, see figure 2.

2/ Surge rating is nonrecurrent and applies only with device in the "on" (conducting) state. The rate of rise of the peak surge current must not exceed 40 amperes/microsecond during the first 10 microseconds after switching from the "off" (blocking) to the "on" (conducting) state. This time is measured from the point where the thyristor voltage has decayed to 90 percent of its initial blocking value.

3/ See figure 2 for maximum current at different conduction angles.

BAROMETRIC PRESSURE, REDUCED: 8 mm Hg at 50 to 300 volts (2N1771A thru 2N1776A)
15 mm Hg at 400 volts (2N1777A)

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1.4.2 Individual ratings.

Types	V_{RM} 1/	V_{FBOM}	V_{RM} (nonrep) 2/
	v(pk)	v(pk)	v(pk)
2N1771A	50	50	75
2N1772A	100	100	150
2N1774A	200	200	300
2N1776A	300	300	400
2N1777A	400	400	500

1/ Values apply for zero or negative gate voltage (V_{GK}).

2/ Peak reverse voltage, nonrepetitive, t = 5.0 ms max.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-202 - Test Methods for electronic and electrical component parts.

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Part I - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20360.)

3. REQUIREMENTS

3.1 General. Requirements for thyristors shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Definitions. For the purpose of this specification, the definitions in MIL-S-19500 and as follows shall apply:

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3.2.1 Turn-off time. Turn-off time is the time required for the gate to regain control of the forward blocking characteristic after forward conduction under stated circuit and temperature conditions. This time is measured from the point where forward current reaches zero to the point of application for forward blocking voltage.

3.2.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in MIL-S-19500 and as follows:

V_{AA} --- Anode power supply voltage (d. c.).

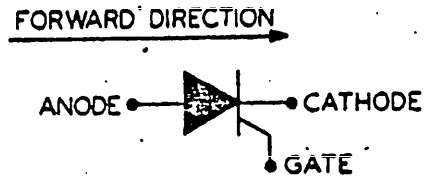
V_{FBX} --- Forward blocking voltage, gate connected to the specified reference terminal.

V_{GG} --- Gate power supply voltage (d. c.).

3.3 Marking. The following marking specified in MIL-S-19500 may be omitted at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

3.3.1 Graphic symbol. The graphic symbol for these devices shall be as shown below.. The polarity shall be indicated by marking the body with the graphical symbol so oriented that the arrow points in the direction of easy flow of the conventional current (forward direction).



3.3.2 "TX" marking. Devices in accordance with the "TX" requirements shall include the additional marking "TX" preceding the type designation.

3.4 Design, construction and physical dimensions. The devices shall be of the design, construction and physical dimensions shown on figure 1.

3.5 Performance characteristics. The performance characteristics shall be as specified in tables I, II, and III.

3.5.1 Process-conditioning, testing, and screening for "TX" types. Process-conditioning, testing and screening, for the "TX" types shall be as specified in 4.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein. The devices covered by this detail specification shall be considered to be structurally similar for the purpose of inspection.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

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4.2.1 Subgroups 1 and 4 of group B inspection and subgroup 6 of group C inspection shall be performed on any structurally-similar type on a sublot basis to qualify that type and all structurally-similar lower-voltage types.

4.2.2 Subgroup 2 of group B inspection shall be performed on a full-sample for each type being qualified.

4.2.3 Subgroup 3 of group B inspection and subgroups 1, 2, 4, and 5 of group C inspection may be performed in accordance with MIL-S-19500 or may be performed on a full-sample of any type to qualify all types.

4.2.4 Subgroup 3 of group C shall be performed on a full-sample for the highest-voltage type to qualify it and all lower-voltage types tested at the same barometric pressure condition.

4.2.5 Qualification testing. The non-TX types shall be used for qualification testing. At the manufacturer's request to the qualifying activity, qualification will be extended to include the "TX" type of the device.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspections. When specified in the contract or order, one copy of the quality conformance inspection data, pertinent to the device inspection lot, shall be supplied with each shipment by the device manufacturer.

4.3.1 Group A inspection. Group A inspection shall consist of examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. Group C inspection shall be conducted on the first lot every six months. Successful completion of the group C inspection shall permit acceptance of lots for the period specified on the basis of completion of group A and group B inspection. If a group C failure occurs, inspection of the failed subgroup shall revert to a lot basis until three-consecutive lots have passed.

4.3.4 Inspection procedure.

4.3.4.1 Subgroups 1 and 2 of group B inspection and subgroup 3 of group C inspection shall be performed on a full-sample size of each type in the lot. Subsequent acceptance of voltage types whi which are different than previously subjected to subgroup 3 of group C within the current specified period, requires retesting of subgroup 3 of group C for a lot including the other voltage type.

4.3.4.2 Subgroup 3 of group B inspection and subgroups 1, 2, 4, and 5 of group C inspection may be performed in accordance with MIL-S-19500 or may be performed on any type within the lot to accept all types in the lot.

4.3.4.3 Subgroup 4 of group B inspection shall be performed on a full-sample size of the highest-voltage type represented in the lot to accept all types in the lot.

4.3.4.4 Subgroup 6 of group C inspection shall be performed on a full-sample size of the highest-voltage type represented in the lot to accept that voltage type and all lower-voltage types for the specified period. Subsequent acceptance of voltage types which are higher than previously subjected to subgroup 6 of group C, within the current specified period, requires retesting of subgroup 6 of group C for a lot including the higher-voltage type. Provisions of MIL-S-19500 early-acceptance procedures do not apply to this specification.

4.3.4.5 Inspection lot. Inspection lot shall be as defined in MIL-S-19500, except that lot accumulation-period requirements shall be six months in lieu of six weeks.

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4.4 Disposition of sample units. Sample units which have been subjected to and have passed subgroups 1, 2, and 4 of group B and subgroup 1, 2, 3, and 6 of group C inspection may be delivered on the contract or order provided that, after the inspection is completed, these sample units are subjected to and pass group A inspection. Subgroup 3 of group B and subgroup 4 and 5 of group C are considered destructive.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, the referenced figures herein, and as follows:

4.5.1 Inspection conditions. All measurements shall be made at T_C = 25 ±3° unless otherwise specified.

4.5.2 End-point mounting torque. The device shall be mounted on a 0.062-inch thick flat copper plate having a mounting-hole diameter of 0.200-inch minimum, 0.275-inch maximum, by means of a #10-32 brass nut. The device shall be held by its hex and the specified torque of 15 pound-inches gradually applied to the nut without shock, using a P.A. Sturtevant model F-100-1 torque wrench or equivalent.

4.5.3 Time limit for end points. End-point tests for qualification and quality conformance inspection shall be completed within 96 hours after completion of the last test in the subgroup.

TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		
	Method	Details	Non TX	TX		Min	Max	Unit
Subgroup 1			10	10				
Visual and mechanical inspection	2071				---	---	---	---
Subgroup 2			5	2				
Reverse blocking current	4211	AC method; bias cond. D; f = 60 Hz			I_{RBOM}			
2N1771A		V_{RM} = 50 v (pk)				---	1.0	ma(pk)
2N1772A		V_{RM} = 100 v (pk)				---	1.0	ma(pk)
2N1774A		V_{RM} = 200 v (pk)				---	1.0	ma(pk)
2N1776A		V_{RM} = 300 v (pk)				---	1.0	ma(pk)
2N1777A		V_{RM} = 400 v (pk)				---	1.0	ma(pk)
Forward blocking current	4206	AC method; bias cond. D; f = 60 Hz			I_{FBOM}			
2N1771A		V_{FBOM} = 50 v (pk)				---	1.0	ma(pk)
2N1772A		V_{FBOM} = 100 v (pk)				---	1.0	ma(pk)
2N1774A		V_{FBOM} = 200 v (pk)				---	1.0	ma(pk)
2N1776A		V_{FBOM} = 300 v (pk)				---	1.0	ma(pk)
2N1777A		V_{FBOM} = 400 v (pk)				---	1.0	ma(pk)

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TABLE I. Group A inspection - Continued

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		
	Method	Details	Non TX	TX		Min	Max	Unit
Subgroup 3			5	3				
Reverse blocking current	4211	AC method; bias cond. D; T_C = 150° C; f = 60 Hz			I_{RBOM}			
2N1771A		V_{RM} = 50 v (pk)				---	2.0	ma(pk)
2N1772A		V_{RM} = 100 v (pk)				---	2.0	ma(pk)
2N1774A		V_{RM} = 200 v (pk)				---	2.0	ma(pk)
2N1776A		V_{RM} = 300 v (pk)				---	2.0	ma(pk)
2N1777A		V_{RM} = 400 v (pk)				---	2.0	ma(pk)
Forward blocking current	4206	AC method; bias cond. D; T_C = 150° C; f = 60 Hz			I_{FBOM}			
2N1771A		V_{FBOM} = 50 v (pk)				---	2.0	ma(pk)
2N1772A		V_{FBOM} = 100 v (pk)				---	2.0	ma(pk)
2N1774A		V_{FBOM} = 200 v (pk)				---	2.0	ma(pk)
2N1776A		V_{FBOM} = 300 v (pk)				---	2.0	ma(pk)
2N1777A		V_{FBOM} = 400 v (pk)				---	2.0	ma(pk)
Exponential rate of voltage rise	4231	Bias cond. D; T_C = 150° C; $\frac{dv}{dt}$ = 5 v/μsec; repetition rate = 60 pps; test duration = 15 sec; C = 1.0 μf; 50 Ω ≤ R_L ≤ 400 Ω			V_{FBO}			
2N1771A		V_{AA} = 50 Vdc				47	---	Vdc
2N1772A		V_{AA} = 100 Vdc				95	---	Vdc
2N1774A		V_{AA} = 200 Vdc				190	---	Vdc
2N1776A		V_{AA} = 300 Vdc				285	---	Vdc
2N1777A		V_{AA} = 400 Vdc				380	---	Vdc
Gate trigger voltage	4221	T_C = 150° C; R_L = 400 Ω; R_e = 20 Ω max			V_{GT}			
2N1771A		V_2 = V_{FBX} = 50 Vdc				0.20	---	Vdc
2N1772A		V_2 = V_{FBX} = 100 Vdc				0.20	---	Vdc
2N1774A		V_2 = V_{FBX} = 200 Vdc				0.20	---	Vdc
2N1776A		V_2 = V_{FBX} = 300 Vdc				0.20	---	Vdc
2N1777A		V_2 = V_{FBX} = 400 Vdc				0.20	---	Vdc
Subgroup 4			5	3				
Gate trigger voltage and current	4221	V_2 = V_{FBX} = 6.0 Vdc; R_L = 50 Ω; R_e = 20 Ω max.			V_{GT} I_{GT}	--- ---	2.0 15.0	Vdc mAdc
Forward "on" voltage	4226	I_{FM} = 14.8 a(pk) (pulse); pulse width = 8.5 msec, max; duty cycle = 2% max			V_{FM}	---	1.85	v(pk)
Holding current	4201	Bias cond. D; V_{AA} = 24 Vdc; I_{F1} = 1 Adc; I_{F2} = 100 mAdc; trigger voltage source = 6 Vdc; trigger PW = 25 μs (min); R_2 = 50 Ω			I_{HOO}	---	25.0	mAdc

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TABLE I. Group A inspection - Continued

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		
	Method	Details	Non TX	TX		Min	Max	Unit
Subgroup 4 - Continued								
Turn-off time	---	(See figures 3a and 3b.) i_{FM} = 5 a(pk); t_p = 100 ±50 µsec; di/dt = 5 a/µsec, min; reverse voltage at t_1 = 2 volts, min; T_C = 145° C; repetition rate = 60 pps, max; dv/dt = 5 v/µsec; gate bias conditions: gate source voltage = 0.0 volts; gate source resistance = 100 ohms			t_{off}			
2N1771A		v_{FBXM} = v_{FBOM} = 50 v (pk); v_{RM} = 50 v, max.				---	30	µsec
2N1772A		v_{FBXM} = v_{FBOM} = 100 v (pk); v_{RM} = 100 v, max.				---	30	µsec
2N1774A		v_{FBXM} = v_{FBOM} = 200 v (pk); v_{RM} = 200 v, max.				---	30	µsec
2N1776A		v_{FBXM} = v_{FBOM} = 300 v (pk); v_{RM} = 300 v, max.				---	30	µsec
2N1777A		v_{FBXM} = v_{FBOM} = 400 v (pk); v_{RM} = 400 v, max.				---	30	µsec
Gate-controlled turn-on time	4223	i_{FM} = 5 Adc; V_{GG} = 10 Vdc; R_g = 50 Ω; t_{p1} = 15 ±5 µsec, 2 a/µsec ≤ $\frac{di}{dt}$ ≤ 200 a/µsec			t_{on}	---	5	µsec
2N1771A		V_{AA} = 50 Vdc						
2N1772A		V_{AA} = 100 Vdc						
2N1774A		V_{AA} = 100 Vdc						
2N1776A		V_{AA} = 100 Vdc						
2N1777A		V_{AA} = 100 Vdc						

TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		
	Method	Details	Non TX	TX		Min	Max	Unit
Subgroup 1			10	10				
Reverse gate current	4219	V_{KG} = 10 Vdc			I_{KG}	---	250	mAdc
Surge current	4066	(See figure 4) i_{FM} (surge) = 60 a; 10 surges at 1 per minute; I_O = 4.7 Adc at rated v_{RM}; T_C = 105° C; f = 60 Hz; surge duration = 7 msec, min.			---	---	---	---

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		
	Method	Details	Non TX	TX		Min	Max	Unit
Subgroup 1 - Continued								
Nonrepetitive peak reverse voltage	---	(See figure 5); 10 pulses			---	---	---	---
2N1771A		V_{RM} = 75 v (pk)						
2N1772A		V_{RM} = 150 v (pk)						
2N1774A		V_{RM} = 300 v (pk)						
2N1776A		V_{RM} = 400 v (pk)						
2N1777A		V_{RM} = 500 v (pk)						
End point: (See 4.5.3)								
Reverse blocking current	4211	AC method; bias cond. D; f = 60 Hz			I_{RBOM}			
2N1771A		V_{RM} = 50 v (pk)				---	1.0	ma(pk)
2N1772A		V_{RM} = 100 v (pk)				---	1.0	ma(pk)
2N1774A		V_{RM} = 200 v (pk)				---	1.0	ma(pk)
2N1776A		V_{RM} = 300 v (pk)				---	1.0	ma(pk)
2N1777A		V_{RM} = 400 v (pk)				---	1.0	ma(pk)
Forward blocking current	4206	AC method; bias cond. D; f = 60 Hz			I_{FBOM}			
2N1771A		V_{FBOM} = 50 v (pk)				---	1.0	ma(pk)
2N1772A		V_{FBOM} = 100 v (pk)				---	1.0	ma(pk)
2N1774A		V_{FBOM} = 200 v (pk)				---	1.0	ma(pk)
2N1776A		V_{FBOM} = 300 v (pk)				---	1.0	ma(pk)
2N1777A		V_{FBOM} = 400 v (pk)				---	1.0	ma(pk)
Reverse blocking current at high temperature	4211	AC method; bias cond. D; T_C = 150° C; f = 60 Hz			I_{RBOM}			
2N1771A		V_{RM} = 50 v (pk)				---	2.0	ma(pk)
2N1772A		V_{RM} = 100 v (pk)				---	2.0	ma(pk)
2N1774A		V_{RM} = 200 v (pk)				---	2.0	ma(pk)
2N1776A		V_{RM} = 300 v (pk)				---	2.0	ma(pk)
2N1777A		V_{RM} = 400 v (pk)				---	2.0	ma(pk)
Forward blocking current at high temperature	4206	AC method; bias cond. D; T_C = 150° C; f = 60 Hz			I_{FBOM}			
2N1771A		V_{FBOM} = 50 v (pk)				---	2.0	ma(pk)
2N1772A		V_{FBOM} = 100 v (pk)				---	2.0	ma(pk)
2N1774A		V_{FBOM} = 200 v (pk)				---	2.0	ma(pk)
2N1776A		V_{FBOM} = 300 v (pk)				---	2.0	ma(pk)
2N1777A		V_{FBOM} = 400 v (pk)				---	2.0	ma(pk)
Gate trigger voltage and current	4221	$V_2 = V_{FBX}$ = 6.0 Vdc; R_L = 50 Ω, R_e = 20 Ω, max			V_{GT} I_{GT}	--- ---	2.0 15.0	Vdc mAdc
Forward "on" voltage	4226	I_{FM} = 14.8 a (pk) (pulse); pulse width = 8.5 msec, max; duty cycle = 2%, max			V_{FM}	---	1.85	v(pk)

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		
	Method	Details	Non TX	TX		Min	Max	Unit
			10	10				
Subgroup 2								
Reverse blocking current	4211	AC method; bias cond. D; T_C = -65° C; f = 60 Hz			I_{RBOM}			
2N1771A		V_{RM} = 50 v (pk)				---	1.0	ma(pk)
2N1772A		V_{RM} = 100 v (pk)				---	1.0	ma(pk)
2N1774A		V_{RM} = 200 v (pk)				---	1.0	ma(pk)
2N1776A		V_{RM} = 300 v (pk)				---	1.0	ma(pk)
2N1777A		V_{RM} = 400 v (pk)				---	1.0	ma(pk)
Forward blocking current	4206	AC method; bias cond. D; T_C = -65° C; f = 60 Hz			I_{FBOM}			
2N1771A		V_{FBOM} = 50 v (pk)				---	1.0	ma(pk)
2N1772A		V_{FBOM} = 100 v (pk)				---	1.0	ma(pk)
2N1774A		V_{FBOM} = 200 v (pk)				---	1.0	ma(pk)
2N1776A		V_{FBOM} = 300 v (pk)				---	1.0	ma(pk)
2N1777A		V_{FBOM} = 400 v (pk)				---	1.0	ma(pk)
Gate trigger voltage and current	4221	$V_2 = V_{FBX}$ = 6.0 Vdc; T_C = -65° C; R_L = 50 Ω, R_e = 20 Ω, max.			V_{GT}	---	2.0	Vdc
					I_{GT}	---	30	mAdc
			15	15				
Subgroup 3								
Thermal shock (temperature cycling)	1051	Test cond. F			---	---	---	---
Thermal shock (glass strain)	1056	Test cond. B			---	---	---	---
Terminal strength (terminal torque)	2036	Test cond. D1; 10 oz-in, 15 sec for cathode terminal; 10 oz-in, 15 sec for gate terminal			---	---	---	---
Moisture resistance	1021	Omit initial conditioning			---	---	---	---
End points: (Same as subgroup 1)								
			10	5				
Subgroup 4								
Blocking-voltage test	---	(See figure 6) T_C = 147° C ±2° C; t = 170 ±24 hours						
2N1771A		$V_{FBOM} = V_{RM}$ = 50 v (pk)			---	---	---	---
2N1772A		$V_{FBOM} = V_{RM}$ = 100 v (pk)			---	---	---	---
2N1774A		$V_{FBOM} = V_{RM}$ = 200 v (pk)			---	---	---	---
2N1776A		$V_{FBOM} = V_{RM}$ = 300 v (pk)			---	---	---	---
2N1777A		$V_{FBOM} = V_{RM}$ = 400 v (pk)			---	---	---	---

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		
	Method	Details	Non TX	TX		Min	Max	Unit
Subgroup 4 - Continued								
End points: (See 4.5.3)								
Reverse blocking current	4211	AC method; bias cond. D; f = 60 Hz			i_{RBOM}			
2N1771A		V_{RM} = 50 v (pk)				---	2.0	ma(pk)
2N1772A		V_{RM} = 100 v (pk)				---	2.0	ma(pk)
2N1774A		V_{RM} = 200 v (pk)				---	2.0	ma(pk)
2N1776A		V_{RM} = 300 v (pk)				---	2.0	ma(pk)
2N1777A		V_{RM} = 400 v (pk)				---	2.0	ma(pk)
Forward blocking current	4206	AC method; bias cond. D; f = 60 Hz			i_{FBOM}			
2N1771A		V_{FBOM} = 50 v (pk)				---	2.0	ma(pk)
2N1772A		V_{FBOM} = 100 v (pk)				---	2.0	ma(pk)
2N1774A		V_{FBOM} = 200 v (pk)				---	2.0	ma(pk)
2N1776A		V_{FBOM} = 300 v (pk)				---	2.0	ma(pk)
2N1777A		V_{FBOM} = 400 v (pk)				---	2.0	ma(pk)
Reverse blocking current at high temperature	4211	AC method; bias cond. D; T_C = 150° C; f = 60 Hz			i_{RBOM}			
2N1771A		V_{RM} = 50 v (pk)				---	4.0	ma(pk)
2N1772A		V_{RM} = 100 v (pk)				---	4.0	ma(pk)
2N1774A		V_{RM} = 200 v (pk)				---	4.0	ma(pk)
2N1776A		V_{RM} = 300 v (pk)				---	4.0	ma(pk)
2N1777A		V_{RM} = 400 v (pk)				---	4.0	ma(pk)
Forward blocking current at high temperature	4206	AC method; bias cond. D; T_C = 150° C; f = 60 Hz			i_{FBOM}			
2N1771A		V_{FBOM} = 50 v (pk)				---	4.0	ma(pk)
2N1772A		V_{FBOM} = 100 v (pk)				---	4.0	ma(pk)
2N1774A		V_{FBOM} = 200 v (pk)				---	4.0	ma(pk)
2N1776A		V_{FBOM} = 300 v (pk)				---	4.0	ma(pk)
2N1777A		V_{FBOM} = 400 v (pk)				---	4.0	ma(pk)
Gage trigger voltage and current	4221	$V_2 = V_{FBX}$ = 6.0 Vdc; R_L = 50 Ω, R_e = 20 Ω, max			V_{GT} I_{GT}	--- ---	2.0 15.0	Vdc mAdc
Forward "on" voltage	4226	i_{FM} = 14.8 a (pk) (pulse); pulse width = 8.5 msec, max; duty cycle = 2%, max			V_{FM}	---	2.0	v(pk)

TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		
	Method	Details	Non TX	TX		Min	Max	Unit
Subgroup 1			20	20				
Physical dimensions	2066	(See figure 1)			---	---	---	---

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TABLE III. Group C inspection - Continued

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		
	Method	Details	Non TX	TX		Min	Max	Unit
Subgroup 2			10	10				
Shock	2016	Nonoperating; 500 G, 1.0 msec; 5 blows each in orientation: X_1, Y_1, and Y_2			---	---	---	---
Vibration, variable frequency	2056	Nonoperating			---	---	---	---
Constant acceleration	2006	5,000 G; in orientations X_1, Y_1, and Y_2 (terminals supported)			---	---	---	---
End points: (See 4.5.3)								
Reverse blocking current	4211	AC method; bias cond. D; f = 60 Hz			i_{RBOM}			
2N1771A		V_{RM} = 50 v (pk)				---	1.0	ma(pk)
2N1772A		V_{RM} = 100 v (pk)				---	1.0	ma(pk)
2N1774A		V_{RM} = 200 v (pk)				---	1.0	ma(pk)
2N1776A		V_{RM} = 300 v (pk)				---	1.0	ma(pk)
2N1777A		V_{RM} = 400 v (pk)				---	1.0	ma(pk)
Forward blocking current	4206	AC method; bias cond. D; f = 60 Hz			i_{FBOM}			
2N1771A		V_{FBOM} = 50 v (pk)				---	1.0	ma(pk)
2N1772A		V_{FBOM} = 100 v (pk)				---	1.0	ma(pk)
2N1774A		V_{FBOM} = 200 v (pk)				---	1.0	ma(pk)
2N1776A		V_{FBOM} = 300 v (pk)				---	1.0	ma(pk)
2N1777A		V_{FBOM} = 400 v (pk)				---	1.0	ma(pk)
Reverse blocking current at high temperature	4211	AC method; bias cond. D; T_C = 150° C; f = 60 Hz			i_{RBOM}			
2N1771A		V_{RM} = 50 v (pk)				---	2.0	ma(pk)
2N1772A		V_{RM} = 100 v (pk)				---	2.0	ma(pk)
2N1774A		V_{RM} = 200 v (pk)				---	2.0	ma(pk)
2N1776A		V_{RM} = 300 v (pk)				---	2.0	ma(pk)
2N1777A		V_{RM} = 400 v (pk)				---	2.0	ma(pk)
Forward blocking current at high temperature	4206	AC method; bias cond. D; T_C = 150° C; f = 60 Hz			i_{FBOM}			
2N1771A		V_{FBOM} = 50 v (pk)				---	2.0	ma(pk)
2N1772A		V_{FBOM} = 100 v (pk)				---	2.0	ma(pk)
2N1774A		V_{FBOM} = 200 v (pk)				---	2.0	ma(pk)
2N1776A		V_{FBOM} = 300 v (pk)				---	2.0	ma(pk)
2N1777A		V_{FBOM} = 400 v (pk)				---	2.0	ma(pk)
Gate trigger voltage and current	4221	V_2 = V_{FBX} = 6.0 Vdc; R_L = 50 Ω, R_e = 20 Ω, max			V_{GT} I_{GT}	--- ---	2.0 15.0	Vdc mAdc
Forward "on" voltage	4226	i_{FM} = 14.8 a (pk) (pulse); pulse width = 8.5 msec, max; duty cycle = 2%, max			V_{FM}	---	1.85	v(pk)

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TABLE III. Group C inspection - Continued

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		
	Method	Details	Non TX	TX		Min	Max	Unit
Subgroup 3			15	15				
Barometric pressure, reduced	1001	t = 60 seconds forward blocking and 60 seconds reverse blocking; pressure: 8 mm Hg, 2N1771A; 8 mm Hg, 2N1772A; 8 mm Hg, 2N1774A; 8 mm Hg, 2N1776A; and 15 mm Hg, 2N1777A						
		Voltage applied during test:						
2N1771A		$V_{FBO} = V_R = 50$ Vdc			---	---	---	---
2N1772A		$V_{FBO} = V_R = 100$ Vdc			---	---	---	---
2N1774A		$V_{FBO} = V_R = 200$ Vdc			---	---	---	---
2N1776A		$V_{FBO} = V_R = 300$ Vdc			---	---	---	---
2N1777A		$V_{FBO} = V_R = 400$ Vdc			---	---	---	---
Subgroup 4			15	15				
Salt atmosphere (corrosion)	1041				---	---	---	---
Subgroup 5			15	15				
Solderability	2026	Cathode and gate terminals, dwell time = 10 ±1 sec (cover flat portion of terminals)			---	---	---	---
Terminal strength (bending stress) (cathode terminal)	2036	Test cond. F, method B; 8 oz; t = 15 ±3 sec			---	---	---	---
Terminal strength (bending stress) (gate terminal)	2036	Test cond. F, method B; 8 oz; t = 15 ±3 sec			---	---	---	---
Terminal strength (stud torque)	2036	Test cond. D_2; 15 lb-in; t = 15 ±3 sec			---	---	---	---
End points: (Same as subgroup 2 except when the 25° C forward and reverse blocking current tests are performed, the device shall be mounted in accordance with 4.5.2)								
Subgroup 6			$\lambda = 20$	$\lambda = 10$				
Intermittent life		(See figure 7); 50 min "on"; 10 min "off"; $T_C = 118 \pm 5°$ C; $I_O = 3$ Adc			---	---	---	---

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TABLE III. Group C inspection - Continued

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		
	Method	Details	Non TX	TX		Min	Max	Unit
Subgroup 6 - Continued								
End point: (See 4.5.3)								
Reverse blocking current	4211	AC method; bias cond. D; f = 60 Hz			i_{RBOM}			
2N1771A		V_{RM} = 50 v (pk)				---	2.0	ma(pk)
2N1772A		V_{RM} = 100 v (pk)				---	2.0	ma(pk)
2N1774A		V_{RM} = 200 v (pk)				---	2.0	ma(pk)
2N1776A		V_{RM} = 300 v (pk)				---	2.0	ma(pk)
2N1777A		V_{RM} = 400 v (pk)				---	2.0	ma(pk)
Forward blocking current	4206	AC method; bias cond. D; f = 60 Hz			i_{FBOM}			
2N1771A		V_{FBOM} = 50 v (pk)				---	2.0	ma(pk)
2N1772A		V_{FBOM} = 100 v (pk)				---	2.0	ma(pk)
2N1774A		V_{FBOM} = 200 v (pk)				---	2.0	ma(pk)
2N1776A		V_{FBOM} = 300 v (pk)				---	2.0	ma(pk)
2N1777A		V_{FBOM} = 400 v (pk)				---	2.0	ma(pk)
Reverse blocking current at high temperature	4211	AC method; bias cond. D; T_C = 150° C; f = 60 Hz			i_{RBOM}			
2N1771A		V_{RM} = 50 v (pk)				---	4.0	ma(pk)
2N1772A		V_{RM} = 100 v (pk)				---	4.0	ma(pk)
2N1774A		V_{RM} = 200 v (pk)				---	4.0	ma(pk)
2N1776A		V_{RM} = 300 v (pk)				---	4.0	ma(pk)
2N1777A		V_{RM} = 400 v (pk)				---	4.0	ma(pk)
Forward blocking current at high temperature	4206	AC method; bias cond. D; T_C = 150° C; f = 60 Hz			i_{FBOM}			
2N1771A		V_{FBOM} = 50 v (pk)				---	4.0	ma(pk)
2N1772A		V_{FBOM} = 100 v (pk)				---	4.0	ma(pk)
2N1774A		V_{FBOM} = 200 v (pk)				---	4.0	ma(pk)
2N1776A		V_{FBOM} = 300 v (pk)				---	4.0	ma(pk)
2N1777A		V_{FBOM} = 400 v (pk)				---	4.0	ma(pk)
Gate trigger voltage and current	4221	V_2 = V_{FBX} = 6.0 Vdc, R_L = 50 Ω, R_e = 20 Ω max			V_{GT}	---	2.0	Vdc
					I_{GT}	---	15	mAdc
Forward "on" voltage	4226	i_{FM} = 14.8 a(pk) (pulse); pulse width = 8.5 msec, max; duty cycle = 2%, max			V_{FM}	---	2.0	v(pk)

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4.6 Process-conditioning, testing, and screening for "TX" types. The procedure for process-conditioning, testing, and screening the "TX" types shall be in accordance with 4.6 through 4.6.6.3 and figure 8. Process-conditioning shall be conducted on 100 percent of the lot prior to submission of the lot to the tests specified in tables I, II, and III. (At the option of the manufacturer, the non-TX type may be subjected to process-conditioning and testing.)

4.6.1 Quality assurance (lot verification). Quality assurance shall keep lot records for three years minimum, monitor for compliance to the prescribed procedures, and observe that satisfactory manufacturing conditions and records on lots are maintained for these devices. The records shall be available for review by the customer at all times. The quality assurance monitoring shall include, but not be limited to: process-conditioning, testing, and screening. (The conditioning and screening tests performed as standard-production tests need not be repeated when these are predesignated and acceptable to the Government as being equal to or more severe than specified herein.)

4.6.2 High-temperature storage. All devices shall be stored for at least 24 hours at a minimum temperature (T_A) of 150° C.

4.6.3 Thermal shock (temperature cycling). All devices shall be subjected to thermal shock (temperature cycling) in accordance with MIL-STD-750, method 1051, test condition F, except that 10 cycles shall be continuously performed and the time at the temperature extremes shall be 15 minutes, minimum.

4.6.4 Acceleration. All devices shall be subjected to acceleration test in accordance with MIL-STD-750, method 2006, with the following exceptions: The test shall be performed one time in the Y_1 orientation only, at a peak level of 5,000 G minimum. The one-minute hold-time requirement shall not apply.

4.6.5 Hermetic seal (fine-leak) test. All devices shall be fine-leak tested in accordance with MIL-STD-202, method 112, test condition C, procedure IIIa or IIIb (using the applicable conditions of 4.6.5.1 or 4.6.5.2).

4.6.5.1 Conditions for procedure IIIa. The devices shall be placed in a sealed chamber and pressurized to 50 psig minimum with helium gas for a minimum of 4 hours. The devices shall then be removed from the chamber and within 30 minutes be subjected to a helium-leak-detection test. Devices shall be rejected that exhibit a leak rate of 5×10^{-7} cubic centimeter (cc) of helium per second when measured at a differential pressure of one atmosphere. All devices exhibiting this leakage rate or greater shall be removed from the lot.

4.6.5.2 Conditions for procedure IIIb. The devices shall be placed in an activation tank which shall be pressurized with Krypton 85 tracer gas in a nitrogen solution for sufficient time to detect a leak-rate of 1×10^{-8} atmospheric cubic centimeters per second (atm cc/sec). Within four hours after pressurization in the Krypton 85 tracer gas, the leak-rate of the devices shall be determined on an attribute basis using the general equation shown below. Any device exhibiting a leak-rate equal to or greater than 1×10^{-8} atm cc/sec shall be removed from the lot. The general equation for use with radioactive-gas leak-test equipment is:

$$Q = \frac{R}{SKT(P_e^2 - P_i^2)}$$

where:

Q = leak rate in atm cc/sec.
R = net counting rate of tested part above background in cts/min.
S = specific activity of the test gas mixture in μCi/atm cc.
K = counting efficiency of the system for the given part in cts/min μCi.
P_e = pressure of test gas in activation tank during pressurization in atm abs.
P_i = pressure inside part under test in atm abs.
T = duration of pressurization in test gas mixture in seconds.

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4.6.5.3 Hermetic seal (gross-leak) test. All devices shall be tested for gross leaks by immersing in noncorrosive ethylene glycol at approximately 100° C for a minimum of 15 seconds and observed for bubbles. All devices that bubble shall be removed from the lot.

4.6.6 Preburn-in tests. The parameters of table IV shall be measured and the data recorded for all devices in the lot. All devices shall be handled or identified such that the delta end points can be determined after the burn-in test. All devices which fail to meet the requirements of table IV shall be removed from the lot and the quantity removed shall be noted on the lot history.

TABLE IV. Burn-in test measurements

Examination or test	MIL-STD-750		Symbol	Limits		
	Method	Details		Min	Max	Unit
Reverse blocking current	4211	AC method; bias cond. D; f = 60 Hz	I_{RBOM}			
2N1771A		V_{RM} = 50 v (pk)		---	1.0	ma(pk)
2N1772A		V_{RM} = 100 v (pk)		---	1.0	ma(pk)
2N1774A		V_{RM} = 200 v (pk)		---	1.0	ma(pk)
2N1776A		V_{RM} = 300 v (pk)		---	1.0	ma(pk)
2N1777A		V_{RM} = 400 v (pk)		---	1.0	ma(pk)
Forward blocking current	4206	AC method; bias cond. D; f = 60 Hz	I_{FBOM}			
2N1771A		V_{FBOM} = 50 v (pk)		---	1.0	ma(pk)
2N1772A		V_{FBOM} = 100 v (pk)		---	1.0	ma(pk)
2N1774A		V_{FBOM} = 200 v (pk)		---	1.0	ma(pk)
2N1776A		V_{FBOM} = 300 v (pk)		---	1.0	ma(pk)
2N1777A		V_{FBOM} = 400 v (pk)		---	1.0	ma(pk)
Gate trigger voltage and current	4221	$V_2 = V_{FBX}$ = 6 Vdc; R_L = 50 Ω, R_e = 20 Ω, max	V_{GT}	---	2.0	Vdc
			I_{GT}	---	15.0	mAdc
Forward "on" voltage	4226	I_{FM} = 14.8a (pk) (pulse); pulse width = 8.5 msec, max; duty cycle = 2% max	V_{FM}	---	1.85	v(pk)

4.6.6.1 Burn-in test. All devices shall be operated for 96 hours, minimum, for the blocking voltage test, using figure 6 and the following conditions:

T_C = 147 ±2° C
2N1771A, V_{FBOM} and V_{RM} = 50 v (pk)
2N1772A, V_{FBOM} and V_{RM} = 100 v (pk)
2N1774A, V_{FBOM} and V_{RM} = 200 v (pk)
2N1776A, V_{FBOM} and V_{RM} = 300 v (pk)
2N1777A, V_{FBOM} and V_{RM} = 400 v (pk)

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MIL-S-19500/168C

Preparing activity:
Navy -EC

Project 5961-0009-16

Code "C"

4.6.6.2 Post burn
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4.6.6.3 Burn-in te
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5. PREPARATION

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6. NOTES

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Navy - EC
Air Force - 11

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Review activities:
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Navy - SH, EC
Air Force - 11, 17,
DSA - ES

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User activities:
Army - SM, M
Navy - CG, MC, AS
Air Force - 19

that exceed the delta (Δ) limits of 4.6.6.2
from the inspection lot and the quantity
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hen the entire lot shall be unacceptable

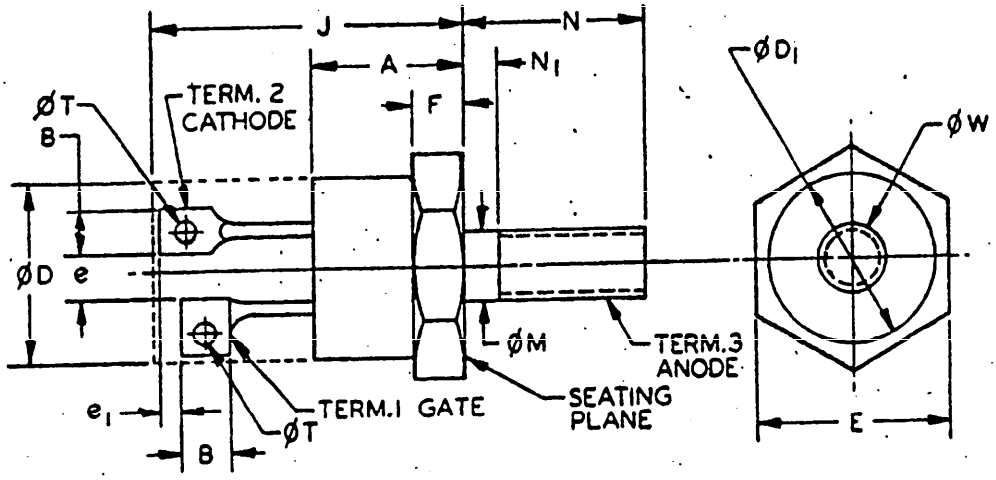
y shall be in accordance with MIL-S-19500.

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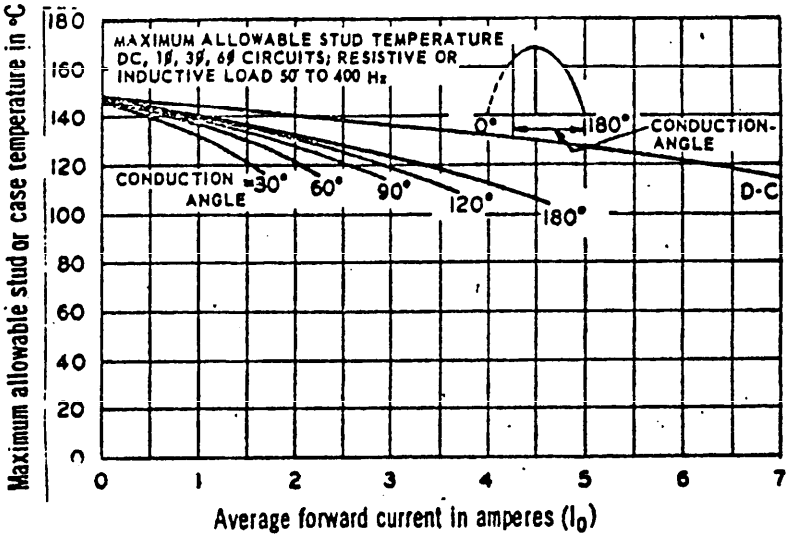


LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.300	.400	7.62	10.16	
B	.080	.136	2.03	3.45	1
ØD	---	.424	---	10.77	2
ØD1	.400	---	10.16	---	3,4
E	.424	.437	10.77	11.10	
e	.013	---	.33	---	7
e1	.060	---	1.52	---	5
F	.060	.175	1.52	4.45	4
J	.700	.855	17.78	21.72	2
ØM	.163	.189	4.14	4.80	
N	.400	.453	10.16	11.51	
N1	---	.078	---	1.98	
ØT	.040	.075	1.02	1.91	
ØW	.1658	.1697	4.212	4.310	6

NOTES:
1. Contour and orientation of fixed terminal lugs are optional.
2. The outline contour (with exception of hexagon) is optional within zone defined by ØD and J.
3. Minimum diameter of seating plane.
4. A chamfer (or undercut) on one or both ends of hexagonal portion is optional.
5. Minimum difference in terminal lengths to establish datum line for numbering terminals.
6. Pitch diameter - thread 10-32 NF-2A (coated). Reference (Screw Thread Standards for Federal Services 1957) Handbook H28.
7. Minimum spacing between terminals.
8. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

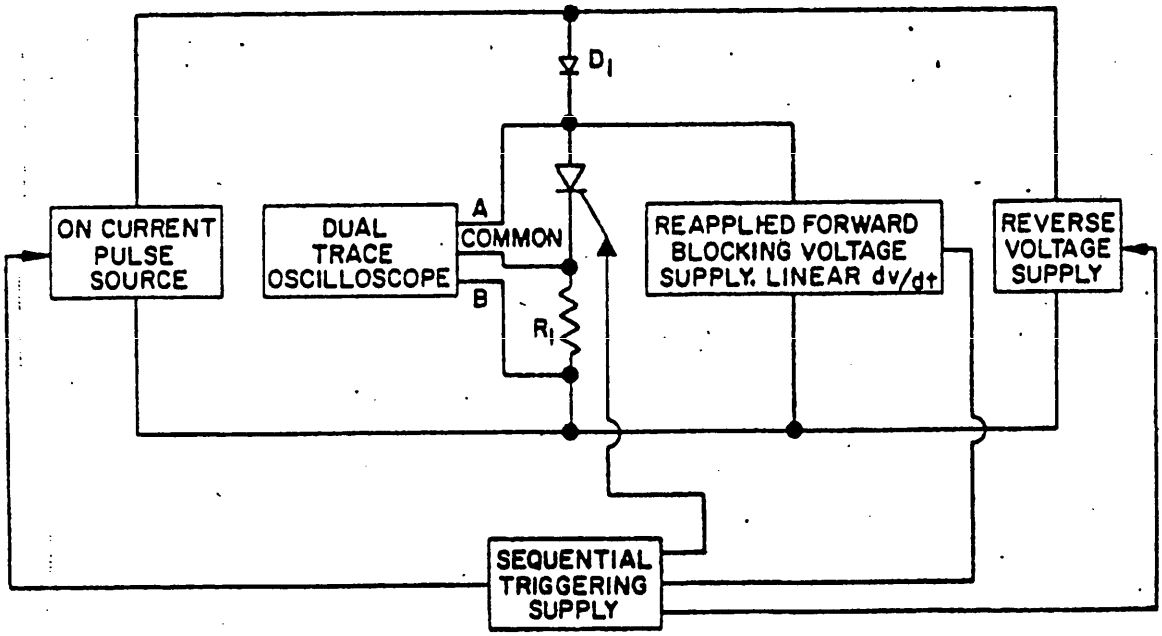
FIGURE 1. Semiconductor device, thyristor, silicon, types (both TX and non-TX) 2N1771A, 2N1772A, 2N1774A, 2N1776A and 2N1777A (TO-64).

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NOTES:
Ratings are derived for .25 watt gate average power dissipation. For higher gate dissipation, decrease stud temperature at rate of 3.1°C. per watt.

FIGURE 2. Maximum operating conditions.



NOTES:
1. R_1 is a non-inductive shunt.
2. D_1 must be selected for recovery time longer than the recovery time of the device under test, but shorter than the turn-off time of the device under test.

FIGURE 3a. Test circuit.

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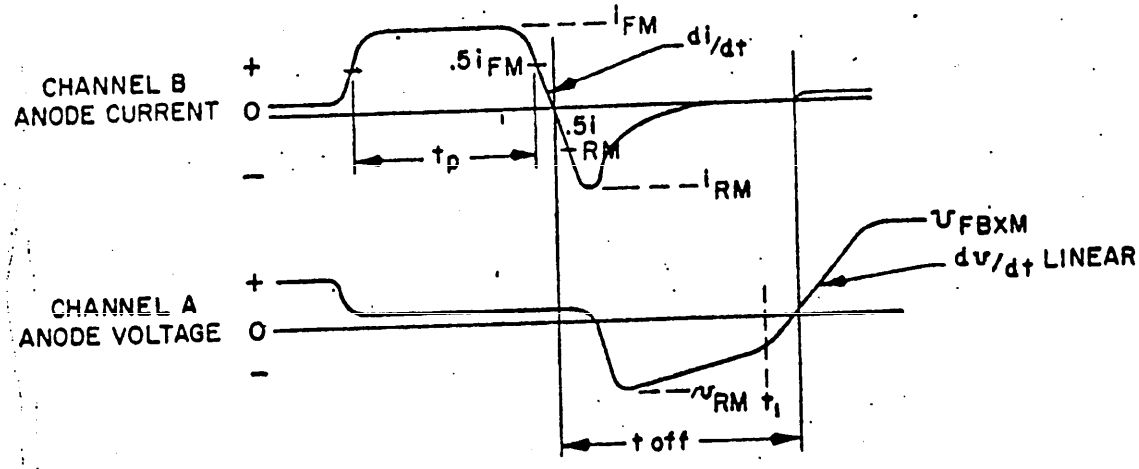


FIGURE 3b. Waveforms.

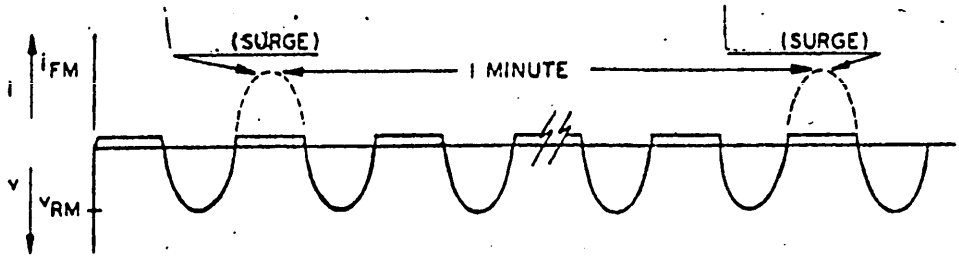
Procedure: The on-current pulse source is adjusted to apply current through the device under test of the specified magnitude, pulse width and duty cycle. The reverse voltage supply is adjusted to apply reverse current to the device under test with the specified di/dt and specified peak reverse voltage and reverse voltage at time t_1. The reapplied forward blocking supply is adjusted to apply forward voltage to the device under test of the specified value with the specified linear rate of rise. The time between when the current passed through zero until the reapplied voltage passes through zero is decreased until either the device under test fails to block the reapplied voltage, or the specified t_{off} is reached. The turn-off time is read from the oscilloscope as shown in figure 3b.

Operation conditions: The devices shall be subjected to the following operational conditions:

a. Forward current amplitude, I_{FM}
b. Forward current duration, t_p
c. Commutation rate, di/dt (the slope of the line from 50% of + peak to 50% of – peak)
d. Peak reverse voltage (max), V_{RM}
e. Reverse voltage at t_1 (min), V_R at t_1
f. Test temperature
g. Rate of rise of reapplied forward voltage, dv/dt
h. Forward blocking voltage, V_{FBXM}
i. Gate bias condition (between gate trigger pulses):
   (1) Gate source voltage
   (2) Gate source resistance.
j. Repetition rate

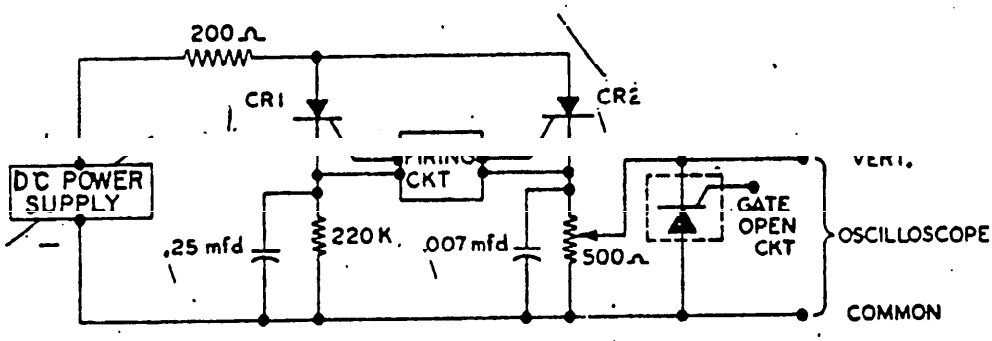
FIGURE 3. Test circuit and waveforms for turn-off time.

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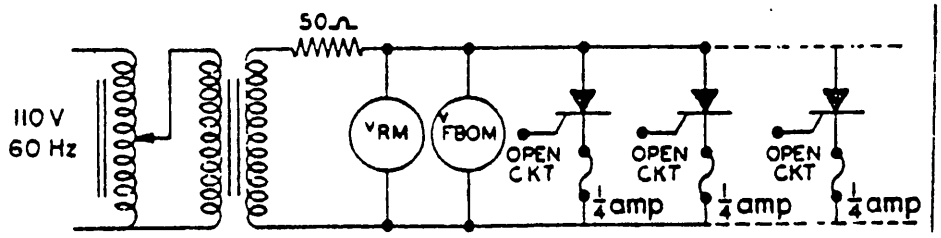
Procedure: The above waveform shall be applied to the devices under test. During this test the gate shall be connected to the anode through a suitable firing resistor and series diode or fired by an equivalent method. The device shall be operating at the specified maximum reverse voltage, specified average forward current and specified case temperature rating in a single phase circuit with a 60 Hz supply and a resistive load. The total peak device current during the surge shall be 60 amps.

FIGURE 4. Surge current waveform.



Procedure: Adjust potentiometer to achieve the specified voltage across the device under test. Timing sequence of firing circuit is: apply trigger pulse to CR2 once every 15 seconds, apply trigger pulse to CR1, 5.0±1.0 milliseconds after firing CR2. The specified number of pulses shall be applied. Alternate circuits shall apply rectangular pulses of 5.0±1.0 milliseconds with a maximum rise and fall time of 50 microseconds.

FIGURE 5. Non-repetitive peak reverse voltage test.

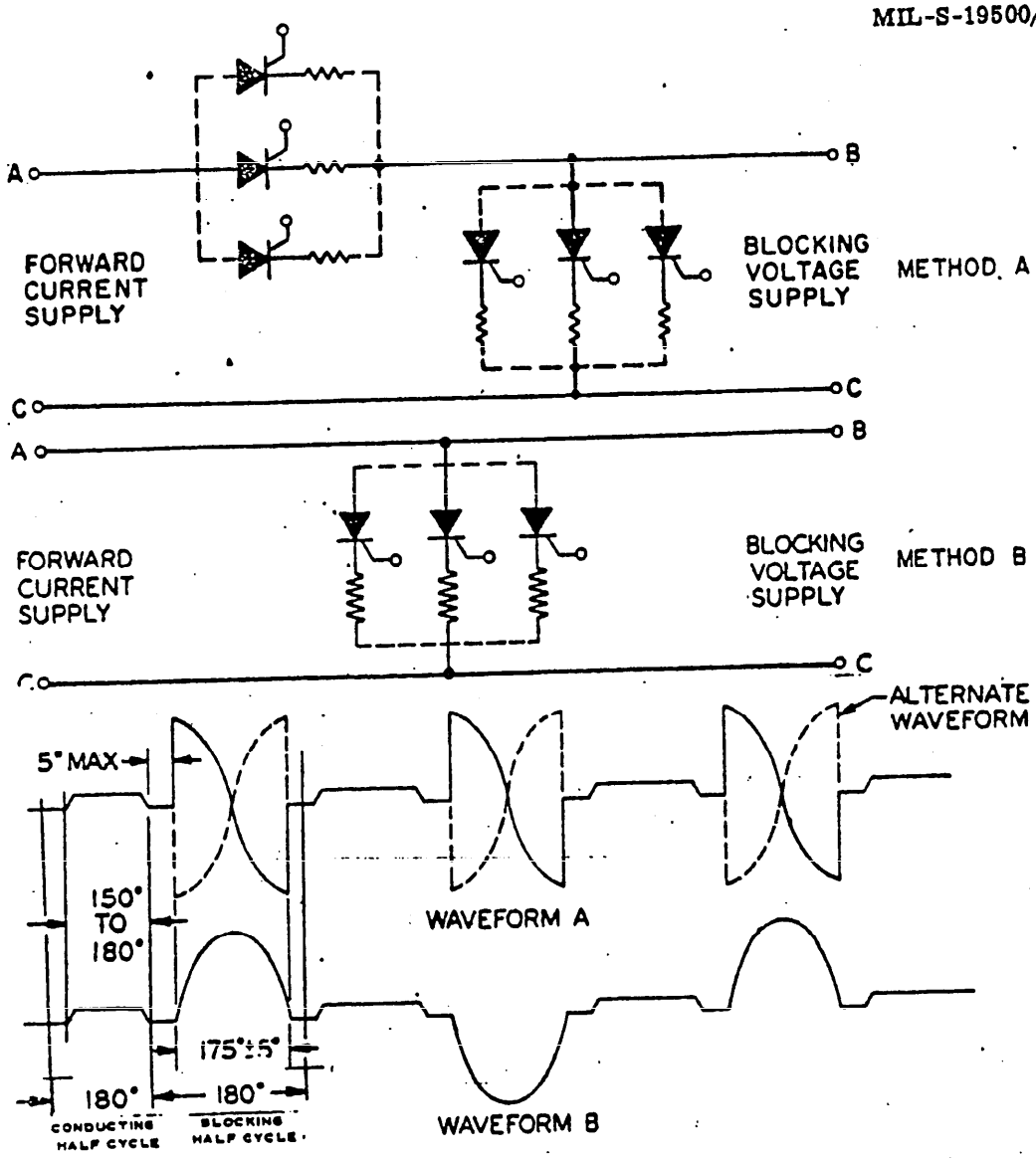


Procedure: Adjust V_{FBOM} and V_{RM} to the specified value for the device under test.

NOTE: Rate of rise of voltage must be limited to values specified in Group A inspection.

FIGURE 6. Blocking-voltage test.

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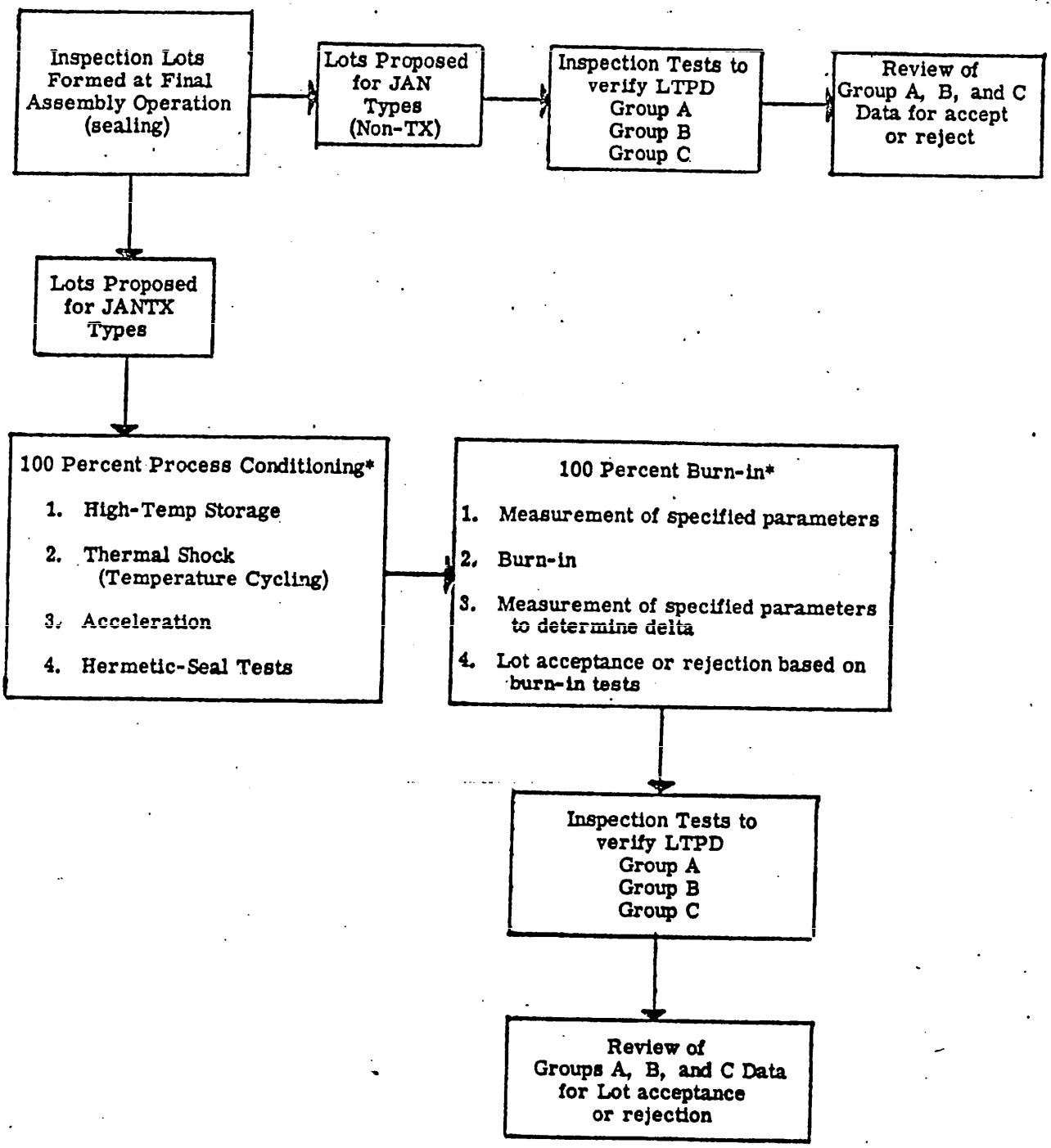
Test circuit details: The test circuit used must provide one of the waveforms shown across the devices. When a large number of devices are to be tested, they may be connected in parallel and/or series combinations with suitable provisions (resistive, reactive, etc. methods) for proper division of forward current or blocking voltages. Suitable protective elements should be used to isolate defective devices without interrupting the test and to protect the remaining devices on test.
The forward current supply shall consist of a low-voltage transformer with a means of adjusting the secondary voltage. It shall supply the specified average forward current to each device under test. The gate supply shall be in phase with the forward current supply. The gate average power input shall not be allowed to exceed .5 watts per device.

Operating conditions: The devices shall be subjected to the following operational conditions:

a. Power sources are to be 60 Hz sinusoidal waveform sources.
b. The conduction angle of I_O shall be 150° to 180°.
c. The rated forward or reverse blocking voltage shall be started during the non-conducting half cycle no later than 5° after conduction has ceased. For waveform A, the RC time constant of the blocking voltage shall be 200±100 microseconds.
d. The duration of the blocking voltages shall be 175±5°.
e. The duration of life test shall be 1000 (+72 –24) hours.

FIGURE 7. Test circuit and waveforms for intermittent life test.

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* ORDER OF THE TESTS IN THE BLOCK SHALL BE PERFORMED AS SHOWN.

FIGURE 8. Order of procedure diagram for JAN (Non-TX) and JANTX types.

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